MIRQAB

AI-Powered Security Operations Platform for SMBs

TEAM

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1 Problem

SMBs face the same phishing, brute-force, ransomware and C2 attacks as large enterprises, but rarely have a staffed Security Operations Center.

- Raw alerts without context
- No owner, no deadline, slow response
- Automation that is either unsafe or absent
- Security logs too sensitive for external AI

2 Solution

One platform that turns security events into **explainable incidents**, gives every HIGH/CRITICAL incident a **5-minute response clock**, and contains endpoints only through an **authorized, agent-confirmed** action — with every step audited.

Deterministic by design; AI stays inside a strict, optional boundary.

3 Target Users

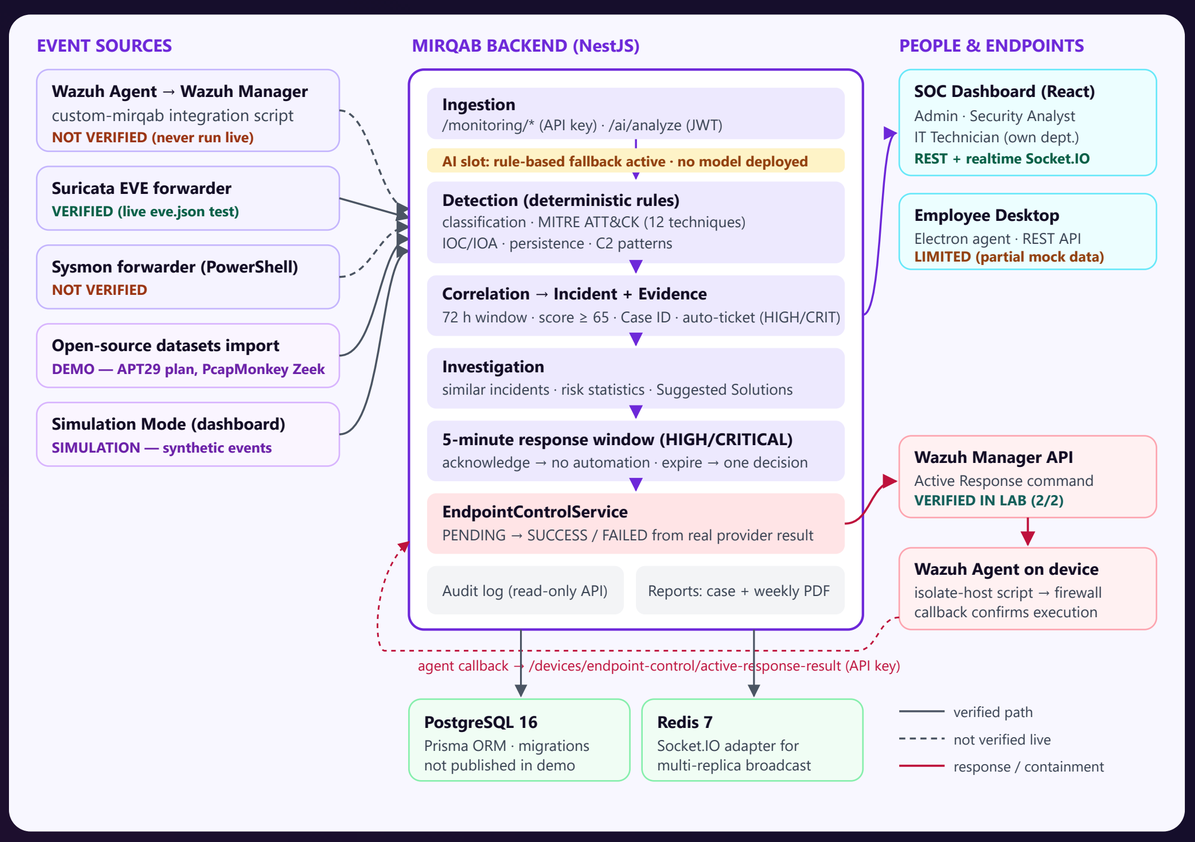
- **Admin** — SMB owner / IT manager
- **Security Analyst** — in-house or outsourced
- **IT Technician** — own department only
- **Employee** — device health via the desktop agent

4 roles · department scoping · organization isolation

4 How It Works — Demo Workflow

Event	Detect	Correlate	Investigate	Respond	Contain & Report
Wazuh · Suricata · Sysmon · datasets · simulation	rules · MITRE ATT&CK · IOC/IOA	72 h window → Case ID + evidence	similar incidents · Suggested Solutions	5-minute window · acknowledge	Wazuh Active Response · PDF reports

5 Architecture (actual)



6 Key Features

- Rule-based detection, **12** curated MITRE techniques VERIFIED
- Correlation into cases, evidence & timeline VERIFIED
- Suggested Solutions from similar past incidents VERIFIED
- 5-minute response window, once-only decision VERIFIED
- Wazuh Active Response isolation VERIFIED IN LAB
- Live network map / Digital Twin VERIFIED
- Case & weekly PDF reports VERIFIED
- Arabic / English, dark / light LIMITED

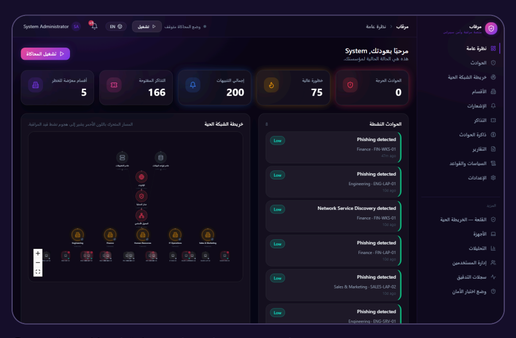
7 Security

- JWT (HS256 pinned) + rotating refresh tokens
- RBAC, department & organization scoping
- Strict input validation, rate limiting
- Authenticated, room-scoped Socket.IO
- Read-only audit log, server-derived actors
- 18-phase security program · **421** backend tests passing (27 Sep 2026)

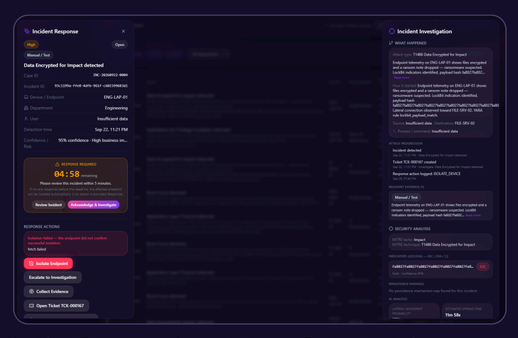
8 Innovation

- **Human-first, time-boxed automation:** containment is decided once, only for HIGH/CRITICAL, only after 5 minutes of silence, never for simulations.
- **Trust only confirmed outcomes:** isolation succeeds only when the endpoint agent confirms it.
- **Private-AI direction:** a self-hosted model slot so logs never leave the organization; AI can advise but never act.
- **Explainable by default:** every indicator, mapping and suggestion traces back to evidence — no match, no claim.

9 Real Screens



Overview (Arabic) — development data



Incident Response + Investigation — 5-minute window

10 Technology Stack

TypeScript · NestJS 10 · React 18 · Vite 5 · Tailwind CSS · Electron 29 · PostgreSQL 16 · Prisma 5 · Redis 7 · Socket.IO 4 · nginx · Docker Compose · Wazuh · Suricata · Jest · Playwright · PDFKit

✓ Honest Status

Working prototype, running locally in Docker NOT DEPLOYED · all analysis rule-based; self-hosted AI NOT VERIFIED · isolation proven on a lab agent only VERIFIED IN LAB · Wazuh GA 4.14.7 BLOCKED · external AI disabled at runtime.